Data Session 7 Key, part 1: Guessing Who People Are

84-355 Democracy's Data | Session 7 — Surnames and BISG | Instructor copy — do not distribute

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All Part 6 code is given to students. Grade the interpretation.

Every number below was run against the data file.

The lab in one sentence

BISG works — well enough that courts rely on it — and it works by attaching a population-level probability to an individual person, which is both its utility and the thing that should never be forgotten about it.

The findings

```r
big <- sn[sn$count > 1000, ]
sapply(grp, function(v) sum(big[[v]] > 90, na.rm = TRUE))
```

```
#>    pctwhite    pctblack      pctapi     pctaian   pct2prace pcthispanic
#>       12022         122         499           6           0        1303
```

```r
rbind(look("WASHINGTON"), look("JEFFERSON"), look("LINCOLN"))
```

```
#>            name rank  count pctwhite pctblack pctapi pctaian pct2prace
#> 145  WASHINGTON  145 177386     5.17    87.53   0.30    0.68      3.78
#> 615   JEFFERSON  615  55179    17.45    74.24   0.40    1.90      3.54
#> 2209    LINCOLN 2209  16477    74.90    14.71   1.35    3.68      2.85
#>      pcthispanic bucket
#> 145         2.54    >5k
#> 615         2.47    >5k
#> 2209        2.51    >5k
```

```r
rbind(SMITH_Allegheny = bisg("SMITH","42003"), SMITH_Bronx = bisg("SMITH","36005"))
```

```
#>                 white black api aian twoplus hispanic
#> SMITH_Allegheny  77.5  19.8 0.3  0.1     1.9      0.3
#> SMITH_Bronx      14.9  73.7 0.5  0.5     1.3      9.0
```

```r
rbind(LEE_Allegheny = bisg("LEE","42003"), LEE_Honolulu = bisg("LEE","15003"))
```

```
#>               white black  api aian twoplus hispanic
#> LEE_Allegheny  48.1  17.1 31.6  0.2     2.7      0.3
#> LEE_Honolulu    3.0   0.9 90.0  0.1     5.7      0.3
```

1. **The asymmetry.** Of surnames held by more than 1,000 people, **12,022** are more than 90% white; **122** are more than 90% Black. A 100-fold gap for a group that is an eighth of the country.
2. **Smith flips.** 70.9% white nationally → **73.7% Black** in the Bronx.
3. **Lee flips the other way.** 42% API by name → 31.6% API in Allegheny, **90%** in Honolulu.
4. **A third of the file is suppressed** — 32.4% of cells, and the Black column is withheld 42.1% of the time among names held by fewer than 200 people. **The gradient is not monotone**: suppression *peaks* at 53.3% in the 500-1k bucket and only then collapses, to 1.7% above five thousand bearers.

The three things to get across

The asymmetry is history, not a data defect. Enslaved people took or were given the surnames of enslavers. Black and white Americans have shared a surname stock for 160 years, which is exactly why surname-only inference fails for Black Americans and works well for recent immigrant groups. Students often assume the method is *biased against* Black Americans in a design sense; it is more precise to say the signal is genuinely absent, for reasons that are themselves the subject of the Voting Rights Act.

Washington and Jefferson are the exceptions, and Lincoln is not. Black share of the name: 87.5%, 74.2%, and 14.7%. Worth pausing on: nobody organised this. It is millions of individual decisions after emancipation, still legible in a 2010 file.

Geography does most of the work. This is the point students most often miss. For ambiguous names — which is most American names — the county is carrying the inference, not the surname. That has an uncomfortable implication: BISG is substantially a method for guessing race from address.

Option A — One name, many places

Verified output

```r
pl <- c("42003","36005","15003","48141","06037","36061")
o <- t(sapply(pl, function(f) bisg("WILLIAMS", f)))
rownames(o) <- c("Allegheny PA","Bronx NY","Honolulu HI","El Paso TX",
                 "Los Angeles CA","Manhattan NY")
o
```

```
#>                 white black api aian twoplus hispanic
#> Allegheny PA     53.2  43.5 0.3  0.1     2.5      0.4
#> Bronx NY          5.5  87.6 0.3  0.3     1.0      5.4
#> Honolulu HI      27.7  18.9 7.0  0.4    43.5      2.4
#> El Paso TX       25.7  39.7 0.3  1.0     3.1     30.3
#> Los Angeles CA  33.5  49.6 1.8  0.4     4.6     10.2
#> Manhattan NY     39.7  52.9 1.0  0.2     2.9      3.3
```

Williams — 45.8% white / 47.7% Black by name alone — ranges across these six counties from **18.9% Black in Honolulu HI** to **87.6% Black in Bronx NY**, with Allegheny at 43.5%. In El Paso it becomes 30.3% Hispanic; in Honolulu it lands at 43.5% two-or-more-races, which is a real feature of Hawaii's population and not an error.

The scale of the problem, now computed in Part 1. For **37.5% of Americans** a surname is nearly decisive — one group above 90%. For **27,497,216 people, about one in 11**, no group reaches even 60%, so the name settles nothing. Williams is the third most common surname in the country, held by 1.63 million people; this is not an edge case.

Say the consequence plainly: BISG is not uniformly good or uniformly bad. It is excellent for Nguyen, useless for Williams, and **the people it fails on are not a random sample of the country** — they are concentrated among exactly the surnames that are shared across Black and white America for historical reasons this course has already covered.

What a good answer says

3/4: Reports the range and identifies which counties move it most.

4/4: Answers the court question honestly. For Williams, nearly all the discrimination between outcomes is coming from the address, because the name itself is a coin flip. A student who says "if I told you only the name you would learn almost nothing, and if I told you only the county you would learn most of what the model outputs" has understood what BISG actually is.

4/4, best: notices the discomfort and states it precisely — the method infers race from residential geography, and residential geography in America is segregated. The estimate is accurate *because* of segregation. That is worth saying out loud, and it is a fair thing for a court to be told.

Common stumble: treating a high probability as a certainty about a named individual. Push on this in feedback every time.

Option B — Which names does geography rescue?

Verified output

```r
for (nm in c("SMITH","WILLIAMS","JOHNSON","LEE","NGUYEN","GARCIA","WASHINGTON")) {
  s <- look(nm)
  before <- max(c(s$pctwhite,s$pctblack,s$pctapi,s$pcthispanic), na.rm = TRUE)
  after  <- max(bisg(nm, "36005"))
  cat(sprintf("%-12s %5.1f%% -> %5.1f%%  (%+.1f)\n", nm, before, after, after - before))
}
```

```
#> SMITH          70.9% ->  73.7%  (+2.8)
#> WILLIAMS       47.7% ->  87.6%  (+39.9)
#> JOHNSON        59.0% ->  82.2%  (+23.2)
#> LEE            42.2% ->  47.8%  (+5.6)
#> NGUYEN         96.5% ->  95.7%  (-0.8)
#> GARCIA         92.0% ->  98.8%  (+6.8)
#> WASHINGTON     87.5% ->  95.4%  (+7.9)
```

In the Bronx the large gains are Williams +39.9 and Johnson +23.2. **Read Smith carefully**: the leading share only moves +2.8, but the *group* it belongs to flips, from 70.9% white nationally to 73.7% Black in the Bronx — the number is nearly unchanged and the answer is completely different. Nguyen (-0.8) and Garcia (+6.8, into an already-decisive 98.8%) are not rescued, because neither needed rescuing.

What a good answer says

3/4: Reports that ambiguous names gain and decisive names do not.

4/4: Explains it as Bayes' rule behaving correctly — a sharp prior is hard to move, a flat prior is easy. Then makes the inference that matters: **the names that need help are the common ones**, and common names cover most Americans. Nguyen and Garcia are precise but describe a minority of the population; Smith, Johnson and Williams are three of the five most common surnames in the country and are all essentially uninformative on their own.

4/4, best: connects to Session 3 (§203). Both labs are about a method that works worst for the population it most needs to serve.

Option C — The cost of suppression

Verified output

```r
sn$bucket <- cut(sn$count, breaks = c(0,200,500,1000,5000,1e7),
                 labels = c("<200","200-500","500-1k","1k-5k",">5k"))
round(100 * tapply(is.na(sn$pctblack), sn$bucket, mean), 1)
```

```
#>    <200 200-500  500-1k   1k-5k     >5k
#>    42.1    49.1    53.3    28.0     1.7
```

```r
c(names_under_1000 = sum(sn$count < 1000),
  share = round(100 * mean(sn$count < 1000), 1))
```

```
#> names_under_1000            share
#>         137364.0             84.7
```

Suppression of the Black column runs **42.1%** for names held by fewer than 200 people, then **rises** to 49.1% and peaks at **53.3% in the 500-1k bucket** before collapsing to 28.0% and 1.7%. **It is not a smooth decline, and the printed table above says so** — do not let the shape be described as one, in your own comments or a student's. **84.7%** of all surnames in the file are held by fewer than 1,000 people.

If a student asks *why* the hump is there, the honest answer is that the file does not say. The Bureau publishes the suppression flag, not the rule that generated it, and published cells include exact zeros — so the pattern cannot be reverse-engineered from what we have. What *is* checkable, and worth showing, is that the peak sits in a different bucket for each column: the Hispanic column is worst at <200, the Black column at 500-1k, the American Indian column at 500-1k. Whatever the rule is, it does not act on how rare the name is alone.

What a good answer says

3/4: Reads the bucket table and notes that most names in the file are rare. Reporting the `<200` figure alone is a 3, not a 4 — it is the headline number and it is not the worst one.

4/4: Answers the neutrality question properly. Suppression is applied by a neutral rule — small cells get hidden — but it does not *land* neutrally. Recent immigrant communities, small national-origin groups, and Native American families are disproportionately the holders of rare surnames. A rule can be neutral in form and unequal in effect, which is a distinction this course has now met three times (the Electoral College, §203's coverage thresholds in Session 3, and here).

4/4, best: proposes what a court should be told — not "the method is unreliable," but specifically that the *precision varies systematically, and not monotonically, with the rarity of the name*, and that any estimate should carry that caveat by subgroup.

Option D — Their own question

Students looking up family names is the most engaged this lab gets. Expect questions about names that "feel wrong"; usually the answer is that the name is rare and half its columns are suppressed.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is your code.
- **The move that earns a 4** is holding utility and limitation together. A paper that concludes "this is racist pseudoscience" and one that concludes "this is a validated method" are both 2s. The method is genuinely relied upon in litigation *and* is genuinely a guess; the assignment is to say both.
- **Watch for individual-level claims.** "This tells us Mr Smith is Black" is wrong and should be corrected explicitly.

Anticipated questions

- *"Isn't this racial profiling?"* — Take the question seriously; it is the right question. The answer is that the alternative is not neutrality but blindness: without some estimate of race, no §2 vote-dilution claim can be brought and no lending discrimination can be detected. The method exists because the records that would answer honestly do not exist. Whether that trade is acceptable is worth ten minutes of class.
- *"Why not just ask people?"* — Some states do collect race on registration. Coverage is partial, non-response is high and non-random, and most states collect nothing. Worth noting which states do.

- *"Is my name's estimate right?"* — For rare names, often not, and the suppressed columns are why.
- *"Does the real method do better?"* — Yes. `wru` adds first names, age, sex, party, and block-level geography. Better in degree, not in kind.
- *"Why didn't we use `wru`?"* — Because the arithmetic is three lines and worth seeing. Also because the package needs a Census API key, and this course does not ask students to obtain credentials.

Connections forward

- **Session 14 (RPV and ecological inference)** is the payoff. BISG estimates *who the voters are*; ecological inference estimates *how groups voted*. Chained together they are the standard evidentiary basis of a §2 case — and the errors compound. Say so explicitly on 1 December.
- **Session 3 (§203)** — both labs are about legal consequences resting on statistical estimates, and both have the same structure: a method least reliable exactly where it matters most.
- **Session 2 (the decennial census)** — the suppression rules students meet here are disclosure avoidance, the same concern that produced differential privacy in the 2020 Census.